

Requirement traceability (ISRO PS-26169)

Every row maps to the module that implements it and the automated test that proves it (npm test; spec tests are in tests/spec.test.mjs).

#	Requirement	Implementation	Test / evidence
1	Screen $\geq$ 2000 $\times$ 2000, optional user-defined	config.js screenSize (clamped $\geq$ 2000), GUI	spec #1 (3000 px screen tracked)
2	Monochrome focal-plane array; colour optional	render.js: mono single-channel frame (default); cameraType: colour renders 3 independently-noisy RGB channels, tracker uses luminance	spec #2 (both modes)
3	Resolution 640 $\times$ 480, optional user-defined	camW, camH	spec #3 (800 $\times$ 600)
4	FOV default 4 $^\circ$ $\times$ 3 $^\circ$, user-defined	fovX, camScale()	spec #4 (FOV 2 $^\circ$ and 8 $^\circ$)
5	Camera update $\geq$ 30 Hz	fixed step dt = 1/fps, fps $\geq$ 30	spec #5
6	Initial camera at screen centre	gimbal.js	spec #6
7	Target type: beacon spot	render.js addBeacon	spec #7-8
8	1 target mandatory, multiple optional	numTargets (decoys)	spec #7-8, decoy test
9	Shape user-defined, default square	square / circle / diamond	spec #9
10	Size 5-20 px, default 10	targetSize	spec #10
11	Initial location user-defined, default random	startMode, startX/Y	spec #11
12	$\geq$ 4 motions; optional spiral, sinusoidal, user-defined	motion.js (7 motions)	spec #12 (all 7 tracked, constant speed verified in engine tests)
13-14	Max pan/tilt 5-10 $^\circ$ /s, default 5	gimbal.js	spec #13-14 (per-frame move never exceeds limit at 5 and 10 $^\circ$ /s)

#	Requirement	Implementation	Test / evidence
15	Update interval ≥ 20 Hz	control computed every frame	spec #15
16	Acquisition ≤ 2 s	wide-view acquisition or camera-only zoom lens (tracker.js, simulation.js)	spec #16-20, tests/zoom.test.mjs, bench A/B (0.6–1.9 s overview; 0.03 s detect / ≤ 1.3 s centred zoom)
17	Tracking error ≤ 10 px	IMM + controller	spec #16-20, bench (0.1–6.3 px; C1 all-disturbances fails: 10.8 px)
18	Target loss $< 5\%$	state machine	spec #16-20, bench
19	Re-acquisition ≤ 1 s	COAST + overview assist	spec #19 (camera knocked off; 0.5 s dropouts)
20	Processing ≥ 20 FPS	ROI detection, 2–8 ms/frame	spec #16-20
21.1	Salt & pepper $\sim 10\%$, Gaussian, Poisson, selectable	render.js addNoise	spec #21.1
21.2	Noise σ up to 20	gaussianSigma	spec #21.2 (measured σ)
21.3	Jitter ± 20 px/frame	simulation.js _jitter	spec #21.3 (bound verified)
21.4	Clear, haze, fog, rain, low light, user-defined reduction	ATMOSPHERES, severity	spec #21.4 (contrast drops, still tracked)
21.5	Platform ± 20 px/frame; linear default; circular, random, spiral, figure-8	makePlatformMotion	spec #21.5 (step bound + tracked for all 5)
—	Turbulence / vibration	render.js, simulation.js	spec turbulence test
—	Endless-run performance log	auto-save every 60 s of simulated time, unload warning	App.jsx
—	Real-time statistics display	App.jsx tiles, plot, minimap	browser run (screenshot)
—	Performance log (duration, FPS, acquisition, avg/max error, lock retention, processing time)	metrics.js JSON/CSV/HTML, auto-save at end of run	spec "performance log" test
—	Benchmark 1: scenarios +	scenario suite, scripts/bench.mjs	RESULTS.md

#	Requirement	Implementation	Test / evidence
	centroiding-error log + auto logs		
—	Benchmark 2: .mp4 @30 fps, PTZ bypassed, centroid error, RMSE, acq/re- acq, lock, FPS	videoTracker.js, videoRunner.js	npm run e2e (H.264 video, ground truth, RMSE 0.07 px)
—	AI-based (title)	trained CNN verifier v4 (74 k parameters) cnn.js / ml/train.py + adaptive IMM estimation	tests/ai.test.mjs (8 tests), npm run ai:eval ablation, report §7
—	Optional demo video (3–5 min)	scripts/make-demo-video.cjs → demo/q- rex-demo.mp4	the recorded file
—	Unseen judge videos (robustness)	tolerant loaders; scripts/video- robustness.mjs (21 videos: 4 resolutions, H.264/MPEG-4, 25/29.97/30 fps, noise types, sizes/shapes/brightness, blink-out, 5 hard cases)	20 of 21 meet the pass line, centroid error 0.01–0.76 px; the 21st (dim, crf 36, 10 % S&P) is informational: 90 % lock, 15 px RMSE
—	Windows / macOS executables verified	desktop/main.cjs --selftest, .github/workflows/ci.yml builds and smoke-tests on real Windows / macOS / Linux runners	self-test passes on Linux; CI run pending the first push
—	Standalone executable	Electron shell (npm run build:exe) and single offline HTML (npm run build:single)	launched on Linux
—	Source code modular and documented	src/engine/*	README, this report
—	Technical report, user manual	docs/TECHNICAL_REPORT.md, docs/USER_MANUAL.md	—